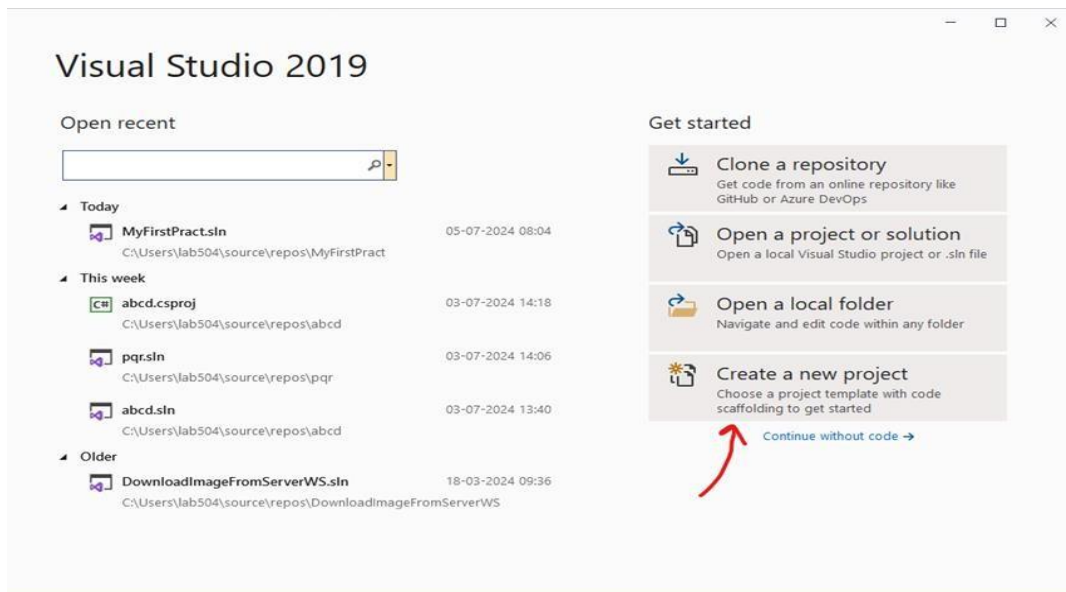


PRACTICAL NO 2

Aim: Setup DirectX 11 , Window Framework and Initialize Direct3D Device

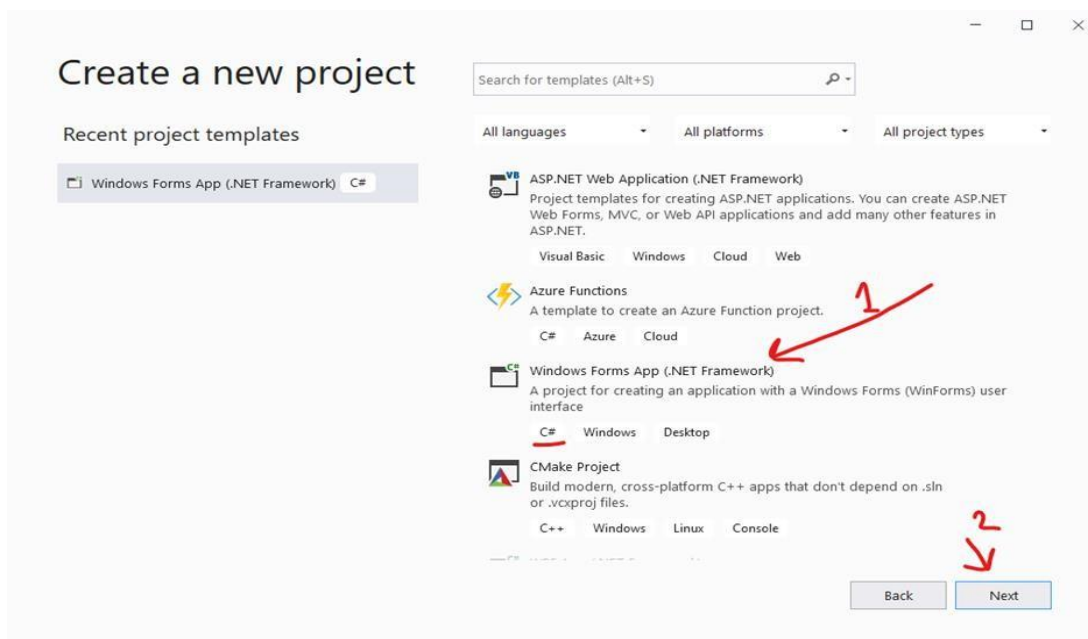
STEP 1:

- Open Visual Studio 2019
- Click on **Create a new project**



STEP 2:

- Select **Window Forms App (.NET Framework)**



Configure your new project

Windows Forms App (.NET Framework) C# Windows Desktop

Project name

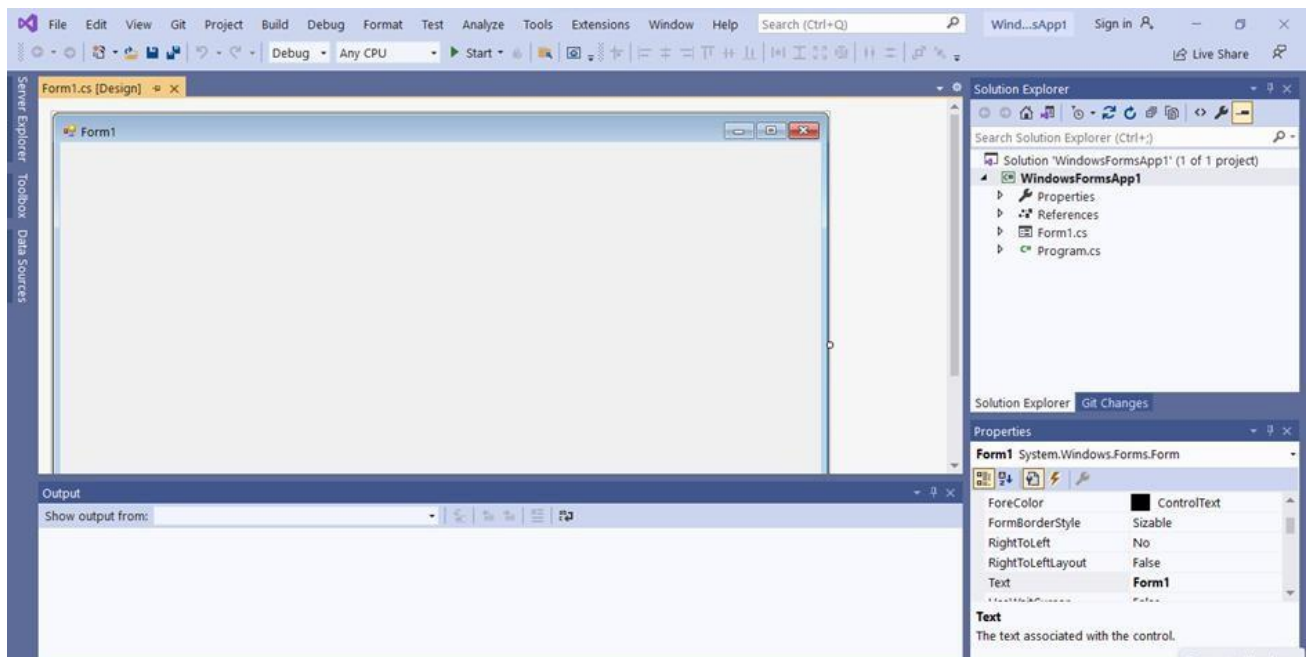
Location

Solution name (i)

☒ Place solution and project in the same directory

Framework

Back Create



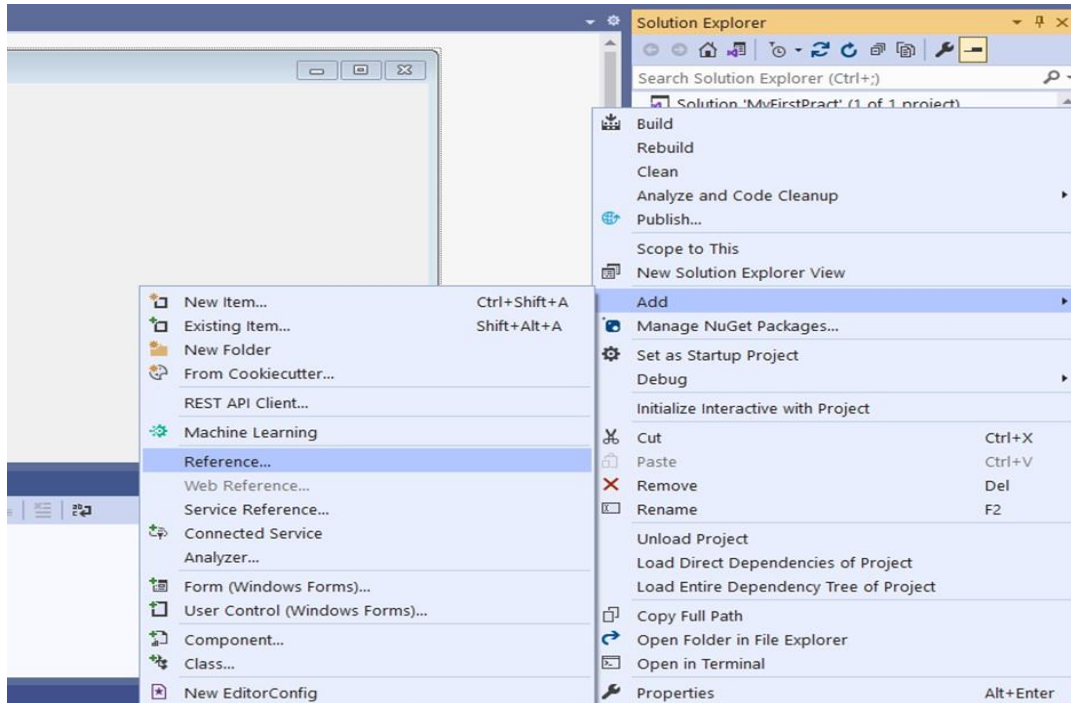
STEP 3 :

- Right click on Project Name
- Click on properties
- Click on Build--> CPU
- Change it from "CPU" to "x86"

STEP 4 :

- DirectX link: <https://www.microsoft.com/en-in/download/details.aspx?id=6812>
- Download
- Now Check whether the DirectX is downloaded or not.

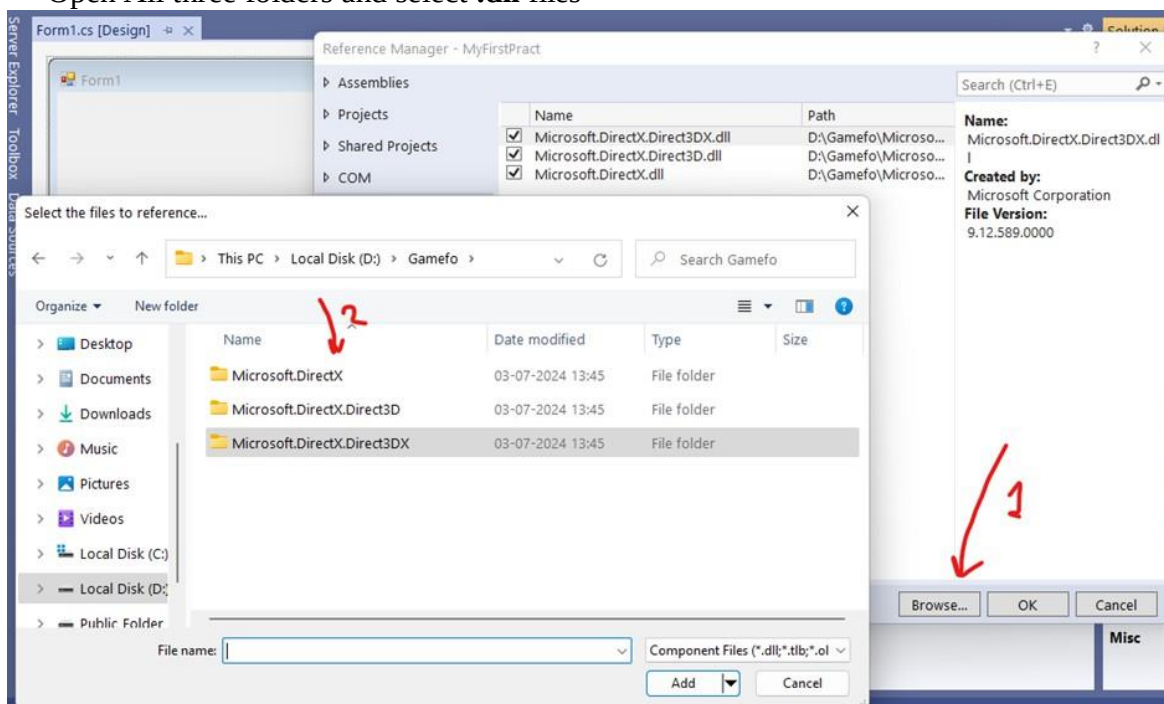
- This PC → Local disk → Windows → Microsoft.NET → DirectX for managed Code → 1.0.2902.0 folder
- click on **Add**
- click on **Reference..**

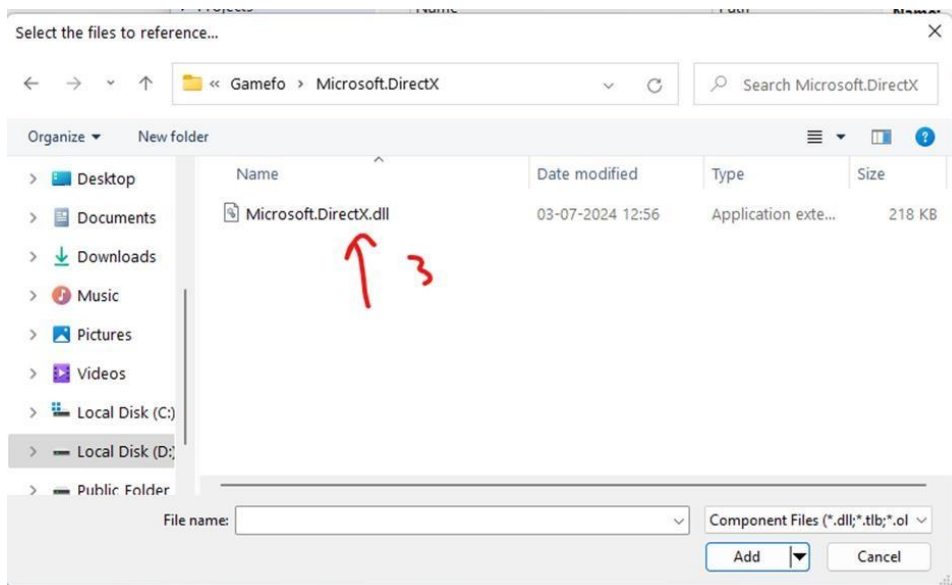


Now Select Open Folders which contains **.dll** files

- **Microsoft.DirectX**
- **Microsoft.DirectX.Direct3D**
- **Microsoft.DirectX.Direct3DX**

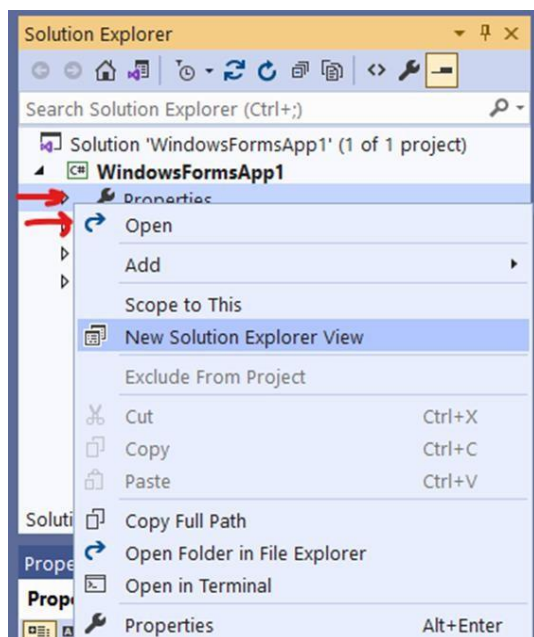
Open All three folders and select **.dll** files



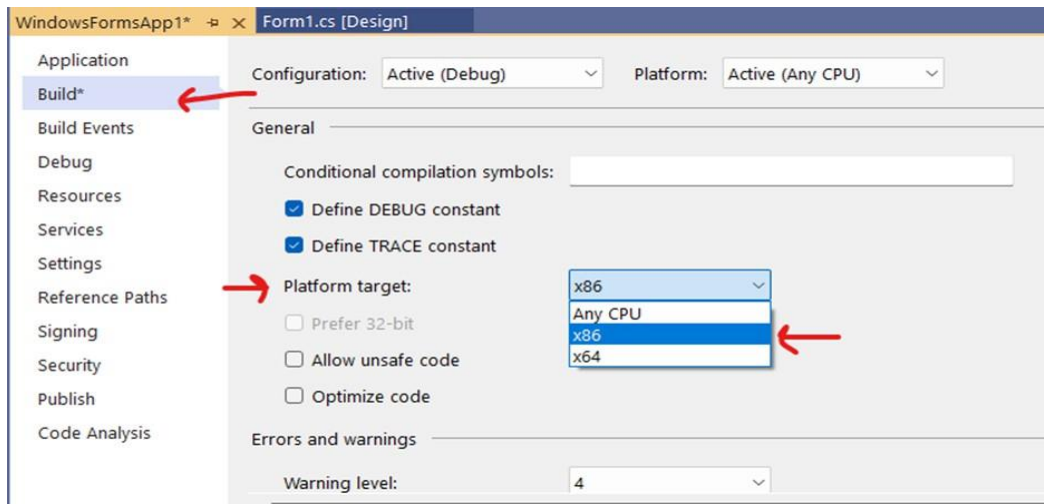


STEP 4 :

- **Right click on Properties**
- Click on **open**

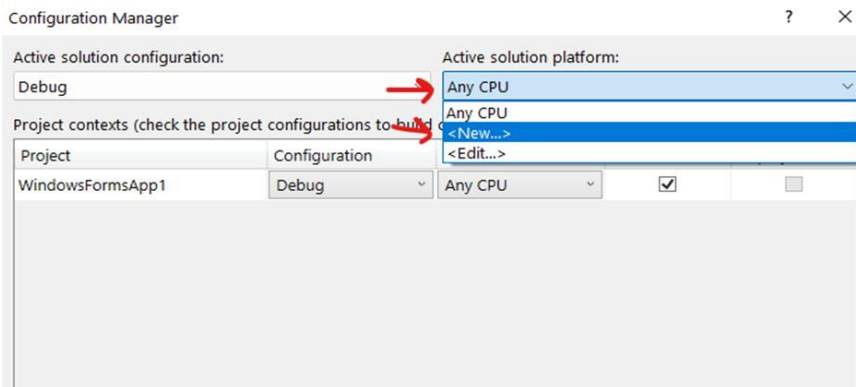
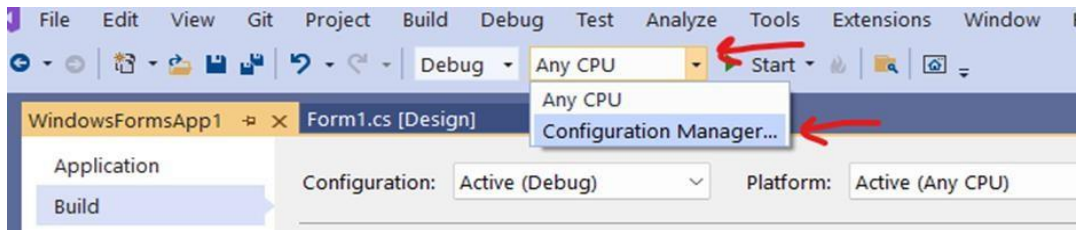


- Click on **Build**
- **Platform target à x86**

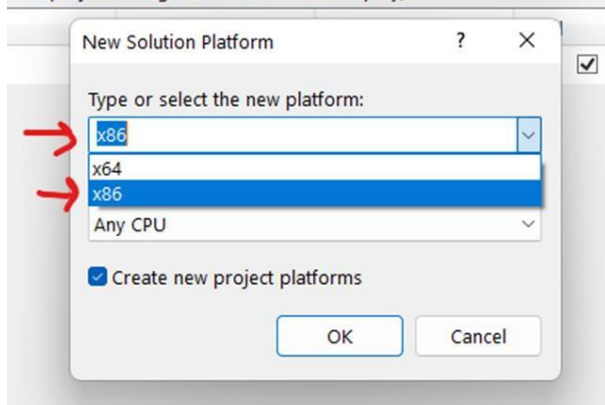


STEP 5 :

- make Any CPU to x86 on **Top menu**

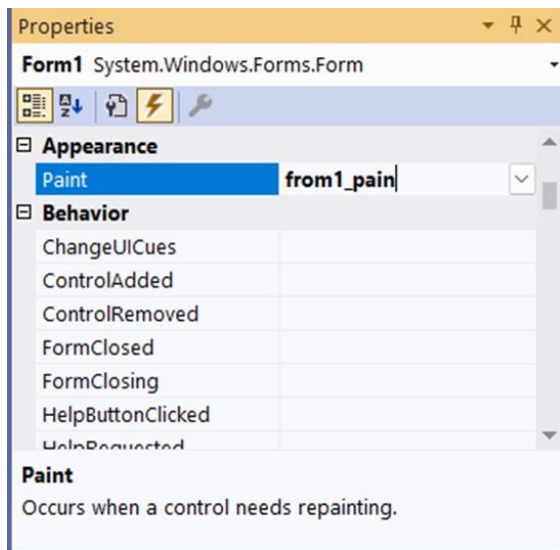


< the project configurations to build or deploy):



STEP 6 :

- **Select file → Events (thunder sign) → Appearance à Paint à name_of_paint**



#PROGRAM :

```
using System;

using System.Collections.Generic;

using System.ComponentModel;

using System.Data;

using System.Drawing;

using System.Linq;

using System.Text;

using System.Windows.Forms;

using Microsoft.DirectX;

using Microsoft.DirectX.Direct3D;

namespace MyFirstPract {

    public partial class Form1 : Form {

        Microsoft.DirectX.Direct3D.Device device;

        public Form1() {
```

```
InitializeComponent();

InitDevice();

}

public void InitDevice() {

PresentParameters pp = new PresentParameters();

pp.Windowed = true;

pp.SwapEffect = SwapEffect.Discard;

device = new Device(0, DeviceType.Hardware, this, CreateFlags.HardwareVertexProcessing,
pp);

}

private void Render() {

device.Clear(ClearFlags.Target, Color.Orange, 0, 1);

device.Present();

}

private void form1_paint(object sender, PaintEventArgs e) {

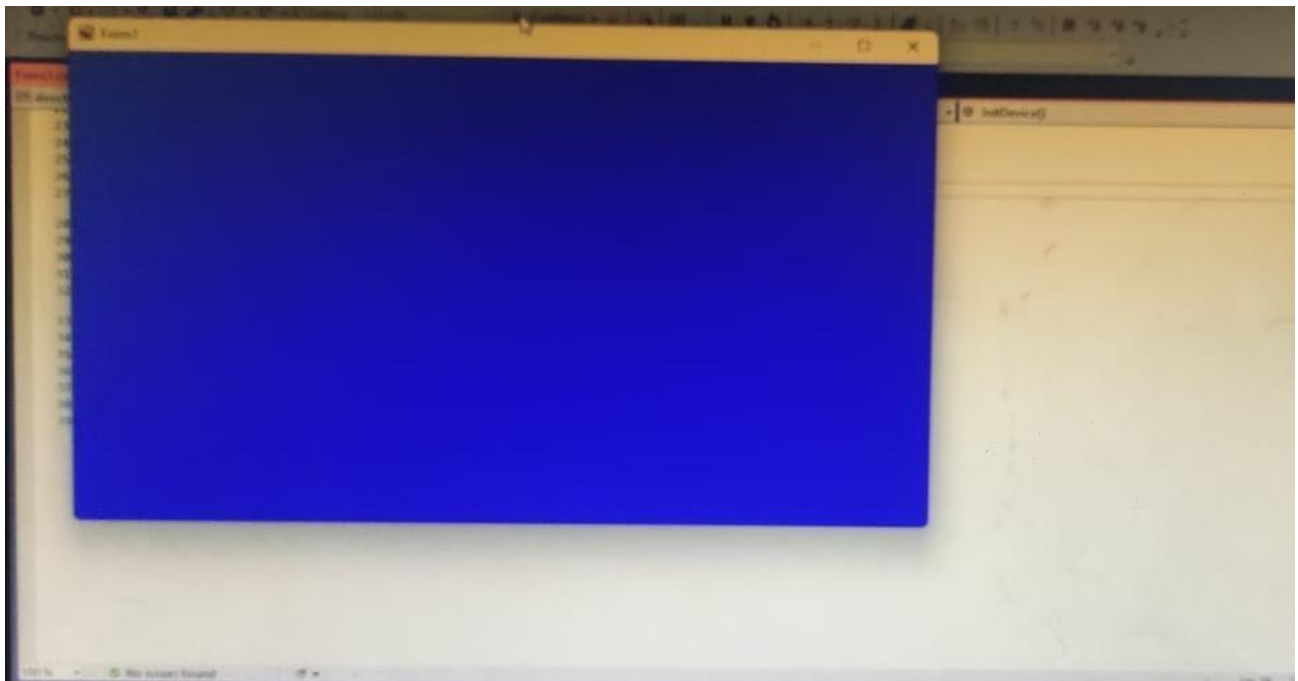
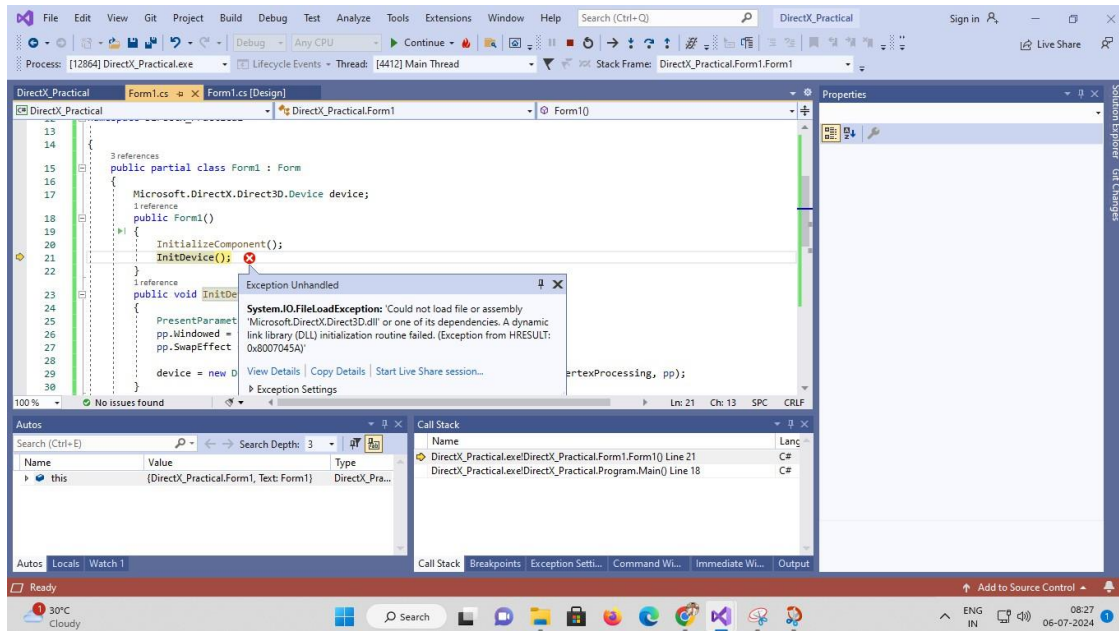
Render();

}

}
```


FINAL OUTPUT :

OUTPUT with (.NET Framework 4) :



OUTPUT with (.NET framework 3.5) :